

If Statement

It is used to execute an instruction or block of instructions only if a condition is fulfilled.

Syntax: -

```
if (condition):
```

```
    statement
```

```
Rest of the Code
```

First the condition is tested, If the condition is True

then the statements given after
colon (:) are executed

```
if (condition):
```

```
    statement1
```

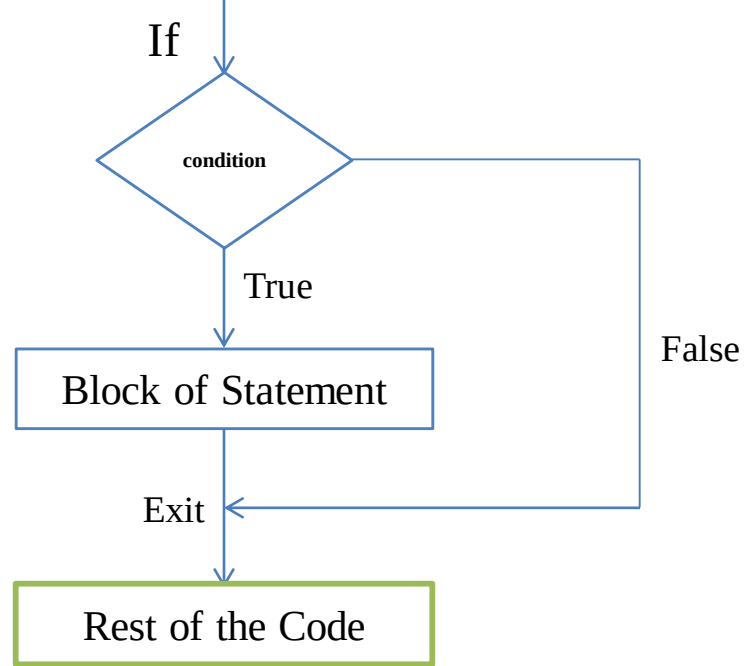
```
    statement2
```

```
Rest of the Code
```

Block of statement/ Group of statements/ Suite

If there is single statement it can be written in one line.

Ex:- if (condition): Statement



Nested If Statement

Syntax :

```
if (condition):  
    block of statements  
    if(condition):  
        block of statements  
    if(condition):  
        block of statements  
if(condition):  
    block of statements  
Rest of the code
```

If Statement with Logical Operator

if ((condition1) and (condition2)):

Statement

Rest of the Code

if ((condition1) and (condition2)):

Block of Statements

Rest of the Code

if ((condition1) or (condition2)):

Statement

Rest of the Code

Indentation

Indentation refers to spaces that are used in the beginning of a statement. By default python puts 4 spaces but it can be changed by programmers.

```
if (condition):  
    Statement  
if(condition):  
    statement 1  
    statement 2  
if(condition):  
    Statement  
if(condition):  
    Statement 1  
    Statement 2  
Rest of the code
```

If Else Statement

if... else statement is used when a different sequence of instructions is to be executed depending on the logical value(True/False) of the condition evaluated.

Syntax: -

if(condition):

 Statement 1

else:

 Statement 2

Rest of the Code

if(condition):

 Statement 1

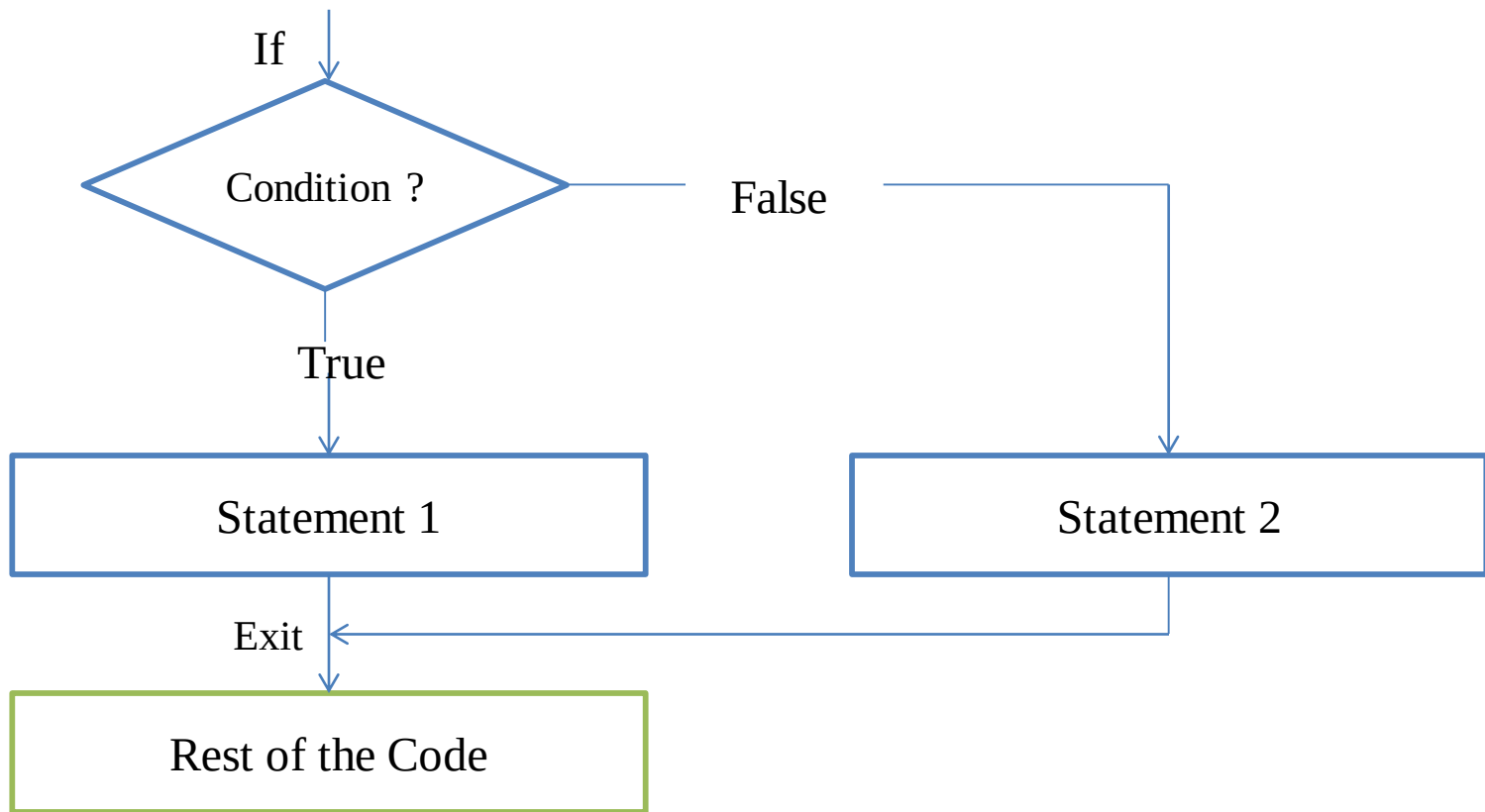
 Statement 2

else:

 Statement 3

 Statement 4

Rest of the Code



Nested If Else Statement

In nested if.... else statement, an entire if... else construct is written within either the body of the if statement or the body of an else statement.

Nested If Else Statement

```
if(condition_1):  
    if(condition_2):  
        Statement 1  
    else:  
        Statement 2  
else:  
    Statement 3  
Rest of the Code
```

```
if(condition_1):  
    if(condition_2):  
        Statement 1  
    else:  
        Statement 2  
else:  
    if(condition_3):  
        Statement 3  
    else:  
        Statement 4  
Rest of the Code
```

if elif Statement

To show a multi-way decision based on several conditions, we use if elif statement.

```
if (condition_1):
```

```
    Statement 1
```

```
elif (condition_2):
```

```
    Statement 2
```

```
elif (condition_3):
```

```
    Statement 3
```

```
elif (condition_n):
```

```
    Statement n
```

```
Rest of the Code
```

if elif else Statement

To show a multi-way decision based on several conditions, we use if elif else statement.

```
if (condition_1):
```

```
    Statement 1
```

```
elif (condition_2):
```

```
    Statement 2
```

```
elif (condition_3):
```

```
    Statement 3
```

```
elif (condition_n):
```

```
    Statement n
```

```
else:
```

```
    Statements x
```

```
Rest of the Code
```

